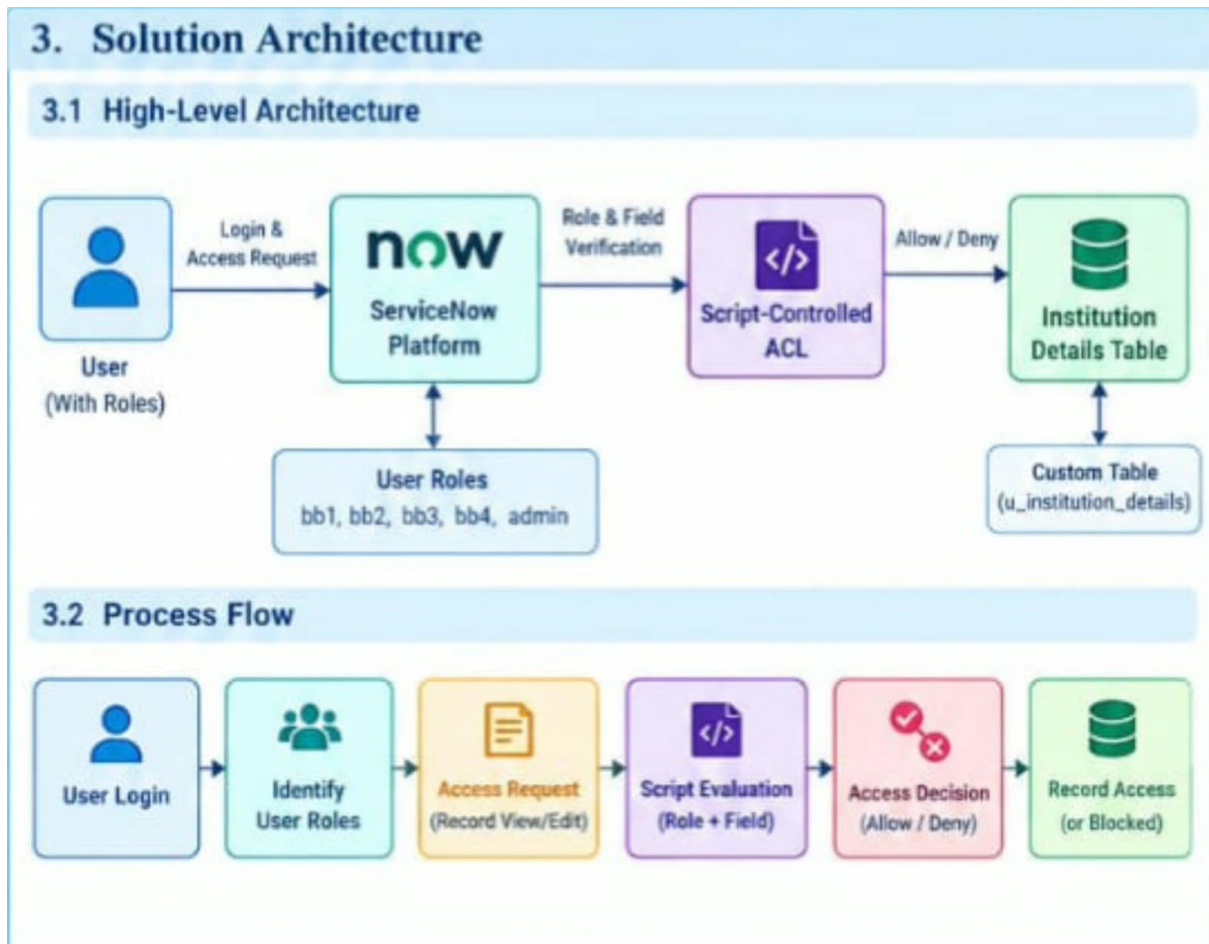


Sub phase-3 solution architecture

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Project name	Script – controlled ACL
Maximum mark	4 mark

3. Solution Architecture

Solution architecture describes the major components of the Script-Controlled ACL system and how they interact with one another.



3.1 High-Level Architecture

The system consists of the following major components:

User

↓

ServiceNow Platform

↓

User & Role Management

↓

Script-Controlled ACL

↓

Access Decision

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Institution Details Table

The user sends an access request through ServiceNow. The system identifies the user's role and evaluates the ACL conditions. Based on the result, access is either granted or denied.

3.2 Architecture Components

Component	Description
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User	Person requesting access to the system
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ServiceNow Platform	Main application platform
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User & Role Management	Maintains users and assigned roles
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Script-Controlled ACL	Evaluates access conditions
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ACL Rules	Define Read, Create, Write and Delete permissions
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Institution Details	Custom table containing institution records
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Database	Stores the application records
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Administrator	Provides unrestricted access and manages the system
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The Institution Details table is defined with fields including Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description.

3.3 Process Flow

The complete system process is:

Step 1 – User Login

The user logs into the ServiceNow platform using valid credentials.

Step 2 – User Identification

The system identifies the logged-in user's roles.

Step 3 – Access Request

The user requests access to a record in the Institution Details table.

Step 4 – ACL Evaluation

The Script-Controlled ACL evaluates the user's role and the required field conditions.

Step 5 – Access Decision

The system determines whether the request should be allowed or denied.

Step 6 – Record Access

If the conditions are satisfied, the user is allowed to perform the permitted operation.

Step 7 – Access Denial

If the user does not satisfy the required conditions, access is denied.

Process Flow Diagram

User Login



Identify User Role



Access Request



Script-Controlled ACL



Role + Field Validation



Access Decision



ACCESS GRANTED

ACCESS DENIED



Institution Details Record

3.4 CRUD Access Architecture

The project separates the different database operations using custom roles.

Operation	Role	Purpose
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Read	bb1	View permitted records
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Create	bb2	Add new records
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Write	bb3	Modify existing records
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Delete	bb4	Remove records
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Full Access	admin	Perform all operations
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This design provides granular control over database operations.

3.5 Institution Details Table

The custom table used in the project is:

Table Label: Institution Details

Table Name: "u_institution_details"

Field Name	Field Type
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Student Roll Number	Auto Number
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Student Name	Reference
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Faculty Name	Reference
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Branch	Choice – ECE, EEE, CSE
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Email	String
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Phone Number	String
--------------	--------

Description| Multi-Line String

3.6 Security Architecture

The security architecture ensures that users receive only the access permitted by their assigned roles and the defined record conditions.

Security Flow

User

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Authentication

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Role Identification

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Script-Controlled ACL

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Field Validation

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Access Decision

↓

Allow / Deny

The design also includes impersonation testing to verify whether users can see only the records permitted to them.

3.7 Expected Result

The expected result of the proposed architecture is:

1. Users with bb1 can view permitted EEE branch records.
2. Users with bb2 can create new records.
3. Users with bb3 can modify existing records.
4. Users with bb4 can delete records.
5. Users without the required roles are denied access.
6. Administrators can access all records.
7. Access decisions are dynamically controlled through ACL rules.

Conclusion

Phase-3 Project Design defines the complete design of the Script-Controlled ACL system. The design combines role-based permissions with field-based access conditions to provide controlled and granular record access. The architecture consists of users, ServiceNow, user roles, Script-Controlled ACLs, and the Institution Details database. By separating CRUD permissions and applying dynamic ACL evaluation, the proposed system provides a structured approach to secure record access.